

BUSINESS ANALYTICS

Group Project Announcement

Customer Churn Prediction & Segmentation for a Digital Bank

An End-to-End Analytics Project: From Raw Data to a Monitored, Deployed Model

Course	Business Analytics
Project Type	Group Project (4–5 students per team)
Domain	Digital Banking — Customer Retention
Deployment Target	Cloud free tier (e.g., Render, Hugging Face Spaces, Streamlit Community Cloud)
Coding Requirement	All code must be written by team members (no no-code/AutoML platforms)

1. Project Overview

Digital banks compete fiercely for customer loyalty in an environment where switching providers is nearly frictionless. This project asks each team to act as the analytics function of a digital bank and answer a concrete business question:

“Which customers are likely to churn in the next quarter, and which segment do they belong to, so that retention efforts can be targeted effectively?”

Teams will build two complementary models on a shared dataset: a classification model that predicts churn risk, and a clustering model that segments customers by behavior. Both models will be engineered, deployed as a working application, and monitored after deployment — mirroring the full lifecycle of an analytics product in industry.

2. Data Sources

Teams must combine the following two data sources:

- **Kaggle dataset (required base):** “Bank Customer Churn Prediction” — <https://www.kaggle.com/datasets/gauravtopre/bank-customer-churn-dataset>. This provides customer demographic and account profile data.
- **Synthetic data (required extension):** Each team must generate additional transactional/behavioral data using Python (e.g., the Faker library plus custom business logic) — monthly transaction history, app login frequency, customer service complaints, channel usage (ATM / mobile app / branch), etc.

This combination ensures the project is not a simple “download-and-train” exercise: teams must justify the realism of the data they generate and document it rigorously.

Data Dictionary Requirement: Every synthetically generated field must be documented in a data dictionary specifying column name, data type, unit, valid range, and the generation logic or business assumption used.

3. Project Structure (8 Modules)

The project is organized into eight modules, suitable for an 8-week timeline (adjustable to fit the course schedule).

Module	Focus Area	Key Tasks
1	Business Understanding & Data Generation	- Define the business problem and key KPIs (churn rate, CLV, retention cost) - Generate supplementary transactional data in Python - Produce a complete data dictionary
2	Exploratory Data Analysis (EDA)	- Analyze distributions, outliers, and correlations - Compare churn rates across customer segments (age, product, balance, etc.) - Visualize behavioral patterns (e.g., RFM-style analysis)
3	Data Cleaning	- Handle missing values, duplicates, and inconsistent categories - Treat outliers (e.g., invalid age, negative balances) - Document all cleaning decisions with before/after comparisons
4	Feature Engineering	- Construct behavioral features (declining transaction trend, login frequency, RFM metrics) - Encode categorical variables and scale numerical features - Address class imbalance (e.g., SMOTE) and perform feature selection
5	Model Development	- Branch A — Classification: Logistic Regression, Random Forest, XGBoost for churn prediction; evaluate with AUC-ROC, Precision/Recall, F1 - Branch B — Segmentation: K-Means or hierarchical clustering on RFM/behavioral features; assign interpretable persona names to each cluster - Integrate both: analyze churn rate by segment, or use segment as a model feature
6	Model Deployment	- Package the model as an API (FastAPI or Flask) - Build a demo interface (Streamlit) simulating a bank-staff lookup tool

Module	Focus Area	Key Tasks
		• Deploy to a cloud free tier (Render, Hugging Face Spaces, or equivalent)
7	Model Monitoring	• Track data drift by comparing incoming data distributions to training data • Monitor performance metrics over time • Build a simple monitoring dashboard (e.g., using the open-source Evidently library) and define retraining triggers
8	Final Report & Presentation	• Translate findings into retention strategy recommendations by segment • Discuss limitations and future improvements • Present to the class with a live or recorded demo

4. Deliverables

Deliverable	Format
Source code (notebooks/scripts)	GitHub repository
Data dictionary	Excel or Markdown file
Deployed API + dashboard demo	Live link (Render / Hugging Face Spaces / Streamlit Cloud)
Monitoring dashboard or report	Notebook or Evidently report
Final report and slides	PDF and PPTX

5. Suggested Team Roles (4–5 members)

Role	Responsibilities
Data Engineer	Synthetic data generation, data dictionary, data cleaning
Data Analyst	Exploratory data analysis and visualization
ML Engineer Classification —	Feature engineering and churn prediction model
ML Engineer Segmentation —	Clustering model and persona definition
MLOps Lead	Deployment and monitoring (may be combined with another role for 4-person teams)

6. Evaluation Focus

Grading will emphasize the rigor and clarity of each stage, rather than model accuracy alone:

- Quality and realism of synthetic data, supported by a complete data dictionary
- Depth of EDA and clarity of insights communicated through visualization
- Soundness of cleaning and feature engineering decisions, with clear documentation
- Appropriateness of model choice and evaluation metrics for the business problem
- A working, accessible deployment (live link required)
- A functioning monitoring component with defined drift/performance triggers
- Originality of code — all code must be authored by team members

7. Submission Logistics

Instructors should insert specific dates for each milestone below before distributing this announcement.

1. Team formation and topic confirmation — [insert date]
2. Module 1–3 checkpoint (data + EDA + cleaning) — [insert date]
3. Module 4–5 checkpoint (features + models) — [insert date]
4. Module 6–7 checkpoint (deployment + monitoring) — [insert date]
5. Final submission and presentation — [insert date]

Questions about scope, data sources, or deployment options should be directed to the course instructor.